CSCI 271: Assignment 2

Name: Marcel Mcgill _____ **Points:** ____ / 25

10 multiple-choice questions covering variables, data types, arithmetic, and input/output from Lesson 2. Circle or check the box next to your answer.

1. What does declaring a variable actually do?

- ☐ A) It immediately sets the value to 0
- ☐ B) It reserves a named location in memory with a specific type
- ☐ C) It prints the variable's name to the screen
- ☐ D) It deletes any variable with the same name

2. Which data type should you use to store a single letter grade, like 'A'?

- ☐ A) int
- ☐ B) string
- ☐ C) char
- ☐ D) double

3. What is the value of int result = 9 / 2;?

- ☐ A) 4.5
- ☐ B) 4
- ☐ C) 5
- ☐ D) It causes a compiler error

4. What does cout << 2 + 3 * 4; print?

- ☐ A) 20
- ☐ B) 9
- ☐ C) 14
- ☐ D) 24

5. What's the difference between << and >>?

- ☐ A) << reads input, >> sends output
- ☐ B) << sends output, >> reads input
- ☐ C) They do the same thing and can be used interchangeably
- ☐ D) << is only for numbers, >> is only for text

6. What does the escape sequence \t do inside a string?

- ☐ A) Starts a new line
- ☐ B) Prints a literal backslash
- ☐ C) Inserts a horizontal tab
- ☐ D) Deletes the previous character

7. Given int x = 5; followed by x = 8;, what happens to the original value 5?

- ☐ A) x now stores both 5 and 8
- ☐ B) x now stores only 8; the 5 is gone
- ☐ C) The program produces a compiler error
- ☐ D) x still stores 5 until the program ends

8. What happens when you write int x = 3.5;?

- ☐ A) It rounds to 4
- ☐ B) It causes a compiler error
- ☐ C) It stores 3, silently dropping the .5
- ☐ D) It stores 3.5 exactly, same as a double would

9. In the line int total = 5;, which word is an identifier rather than a keyword?

- ☐ A) int
- ☐ B) total
- ☐ C) double
- ☐ D) char

10. Why is x = x + 1; valid in C++, even though it looks wrong in math class?

- ☐ A) It's a math error — x can never equal x + 1
- ☐ B) It updates x by taking its current value, adding 1, and storing the result back in x
- ☐ C) It compares x to x + 1 and returns true or false
- ☐ D) It only works if x was never given a value before

Mini Lab: Try It Yourself

```
// Your task:
// 1. Ask the user to enter their name and age
// 2. Print a greeting using both values

// Example output:
// Hello, I am Choco! I am 1 year old.
```

Write the C++ program (explain all the lines) and add the screenshot of your output:

#include <iostream> // allows the use of input and output (cin/cout)

#include<string> // allows me to define string variables

```
using namespace std; // QoL, removes the need to write ::std before everything



int main() // Start Point
{
string name=""; // Empty name variable, primed for definition, has to come before input
cout<<"Please enter your name.\n"; // Displays text with line break
cin >> name ; // Calls for input, stores it in the name string variable
cout << name <<"... thats a nice name."; // Displays the name variable followed by text
cout << " How old are you "<< name<< "?\n"; //Displays text and name variable, and inserts a line break
string age=""; //  empty age string variable, could’ve done this with int but string was easier
cin >> age; // Calls for input to fill the age variable
cout<< age<<" huh..\n"<< name<<" you're pretty olddd!"; //displays age variable, inserts line breaks, and prints text
return(0); //End point

}
```